

IAC

Monday, August 31, 2026 9:55 AM

What do you understand by IAC. Infra Structure as Code.

We have more than 10000 application running in the cloud for application we deploy in azure and aws for data analytics it goes to gcp.

As it is a retail chain. We need frequent update changes as per the industry standard.

We need to fullfill some complaince

1. HIPAA: CVS Store using the same they selling medicine
2. PCI DSS:

All the application deployment need to be done without the support of the operation team.

Frequent deployment in production. Is required.

A developer modify the application till they deploy there is no way they will know how the application works.

We found cloud is the best solution available. We need it we can destroy the infra.

We need a common tool to do the deployment

1. Automation tool aws has cloud formation template cft
2. Arm template
3. Gcp can generate automatic terraform of json template.

A single automation tool can manage all this cloud service provider easily.

It is not programming. Down the line terraform use go lang

Terraform use something called a wrapper language. Which is simple english.

Already the definition is available in terraform registry you just need to make a little bit of changes and use the same.

You need to understand how cloud works.

1. I want to launch an vm in the cloud.

What is the first thing we need to do.

Once you get a laptop in your home

Connect to wifi. Which will give me networking.

Without the network we cannot launch a server in the cloud

Create a vpc

Create a subnet

Inside the subnet we need to provide internet connection

Then only I can launch an server in the cloud.

The isp in your home give you internet connection

You need to get a router plugin the internet connection

Inside the router you need to configure the lan ip address scheme

Then only your all the machine will be connected and shared the internet connection.

Vpc

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First we will learn how to create a vpc. Virtual private cloud.

Region: what is the purpose of this region.

1. Latency
2. Government norms. Human should not be stored outside the country.

Muskan is raj. Base location will be banglore.

Once you go to banglore.

You will take an flat

To work you need to have internet connection

In you home you need an ISP

They will give you just an internet cable

Put is inside your router and create your own custom network. Security

VPC is all about creating a custom network.

You have seen the region inside the region there will be mutiple application running. 10k app.

Every application required custom networking.

Create a isolated network for you.

Network

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We created class b network /16
Subnetting.

8 bits it can process 8 bit of information 1 and 0
16 32 64 128 256

So your network are divided into three classes

Class a

255.0.0.0

Class b

10.0.0.0 till 255 : 255 devices

11.0.0.0 till 255

12.0.0.0 till 255

13

14

255.255.0.0

11111111.11111111.00000000.00000000

8+8=16

10.0.0.0/16- allow me to attach 65devices

We can attach 65565 device

Class c

192.168.0.0 till 255: in class a we can attach 255 devices

192.168.0.1

255.255.255.0

11111111.11111111.11111111.00000000

8+8+8=24

192.168.0.0/24

Cloud console

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Same activity you are going to do it using terraform.

1. Dev test staging an prod environment for my app. Ecommerce app.

There are frequent changes happen in my application.

Frequent deployment.

It will be a better idea that we created a template and using the template we deploy our resources in the cloud.

By mistake ig has been attached with our db server.

There are plenty of work which happen in duplicate with small changes.

Dispatch module

Logistics module

Which required similar kind of infra.

Change some of the parmeter and deploy.

In walmart we hire every month 30 people in software development team.

I need to have a template which will automatically putting the user name automatically will be added to my activity with required permission.

You need to find out repetitive task.

I can easily automate the same using terraform

sta

Stability

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I create a infra using terraform. Well secure architecture for my application .

One of the finance guy goes through aws foundation course.

He find that if he put some tags in the resource in the aws cloud. He can generate budget report.

He didn't bother to inform you.,

You got an new change request do terraform apply all the tags is gone.

Bajaj finance. They run there resource in azure cloud.

Finance company in india. Pinchpenny. Stingy with money.

They want to control the cloud cost. They have added tags. In the active directory. Any user who do not put the tags in the resource automatically after 15 min it will be deleted. They keep on changing the tags.

I build the environment using terraform someone change the environment outside terraform this concept in terraform is called as drift.

How to manage this drift

Ifra failure.

Db server created using terraform db guy who change the storage size 50 100 gb.

Terraform apply everything destroyed and recreated

You build using terraform a log file create which contains your infra

Cloud service provider console

Terraform apply it will read your log file compare current with desired state show mismatch

Previous the state compare with the current state desired state. If there is an mismatch it show you Yes

Terrform will destroy and recreated

Code and apply using terraform template

Using terraform we write down the code

You want to scan the code for vulnerability

22=0.0.0.0/0

3389=0.0.0.0/0

Checkov will show this as valnurabilty

Terraform installation

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In aws cloud we will create an windows machine

Login to the windows machine

Inside windows machine

1. Terraform
2. Git
3. Vscode
4. Aws cli

Think that windows machine is your workstation from where you are going to run your terraform script

But I need to install those packages in my workstation.

We have an package manager called as choco

It is similar to yum or apt in your linux system

We will install choco inside my windows machine

Terraform init

Terraform validate

Terraform plan

Terraform destroy

###all this command will execute inside current directory

Terraform workflow

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Terraform init: it will download the plugin required to communicate with your cloud service provider

Terraform validate: it will only validate your file for syntax

Terraform plan: read all your manifest file from current directory give you and desired state by comparing those manifest file with the cloud service provider

Terraform apply: read all your manifest file--current state--provide an desired state--yes-it will create the infra-it will store the infra in a file statefile. In json format

#finall we need to destroy the resources: to destroy the resources terraform will read which file

Json

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Java notation.

1. I want to create an vpc subnet
{vpc}
{subnet}

Vpc {
Name: gopal-vpc argument-identifier
Cidr: 10.0.0.0/16
//comment

/*
Multip
*/

}

Demo1

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1. We will create a folder in our rdp
2. Then open visual studio code and select the folder
3. Inside the folder we will create three files
4. Provider.tf
5. Vpc.tf

Recap

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1. IAC : infra structure as code. How it is going to help you build your infra when there is a need and you can destroy it also when there is no need
2. Terraform workflow
 - a. Terraform init: it is used to download the provider as per the provider specification. Cloud
 - b. Terraform validate: which will validate your syntax
 - c. Terraform plan: this is going read all your manifest file which si like an current state compare the same with cloud serivice provider and give you an desired state. Dry run
 - d. Terraform apply: which will also give you current state with desired state yes it will make the changes in the cloud service provider and store those changes inside file called as tfstate
 - e. Terraform destroy: destroy the resource it need to read a file tfstate file and destroy the resources.

All this command will execute inside the current directory.

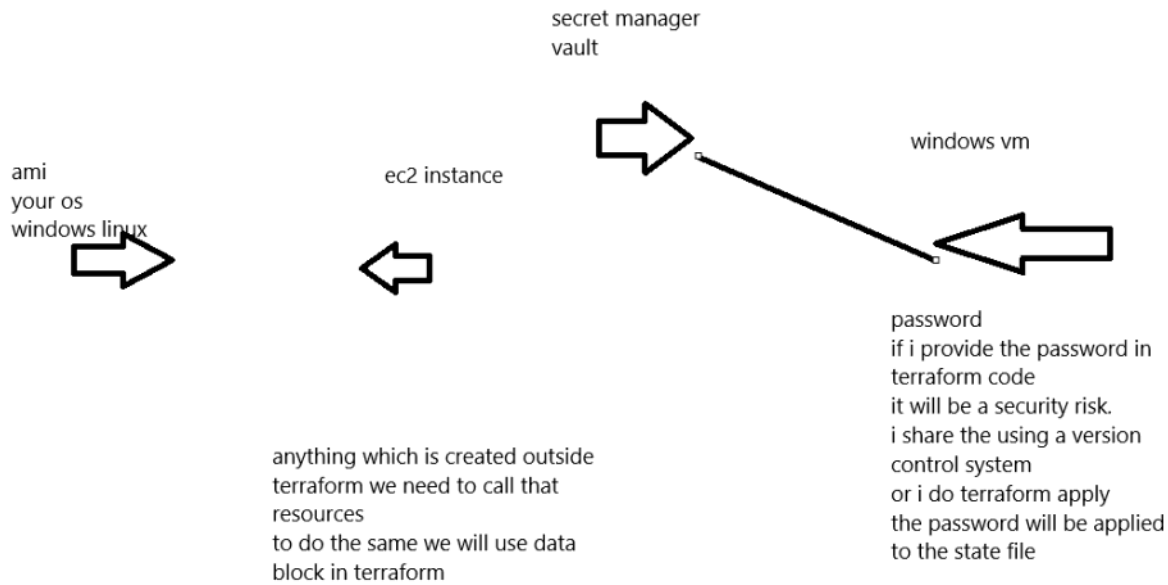
Resource block

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The first block in terraform.

The purpose of the resource block as its name suggest it will going to create resources

1. We will create resources. I want to create vpc subnet ig routetables rds eb container
 2. Call an existing resources: why we need to call an existing resources.
- When we launch an ec2 instance we call some resources. ami



Screen clipping taken: 9/1/2026 9:53 AM

Resource block

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Resource block is divided into three parts

Resource	Nameoftheresource	Refrenceblock
----------	-------------------	---------------

Inside the resource block we provide argument/identifier in key value format

1. We have created vpc
2. Inside vpc we provide argument like a name and address block

Vpc: how to create the vpc

Subnet: how to create subnet

Internetgw: how to create internetgw

Customrtb: how to create customrtb

Attach ig with custom rtb: how to attach ig with custom rtb

Attach app subnet with custom rtb: explicilty attach app subnet with rtb

How to launch an instance with custom vpc and subnet

Then we launch the instance

1. Image size networking custom vpc subnet sg after that only we luanch the instance and have access to the instance

There are two purpose of the refrence block

1. All the resource defination will be stored inside your state file using the refrence block
2. Mapping of resources we use refrence block

Inside your current folder the refrence block cannot be duplicated for the same resource type

Variables

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1. Input variables
2. Output variables
3. Local block

Variables: once declared I can reuse
Make my code much more readable.

The vpc reference is used in multiple places I can call it in different places
If I build a terraform code which is going to create some resource in AWS cloud.
By modifying some of the variables we can create similar kind of resource in different region or it can be used for different projects.
I want to change only VM size using variables I can easily modify the VM size
I have created security group and I just want to add a new port number

Gopal

Happiness: small wrinkle
Sadness: raised inner corner of the eyebrows
Fear: stretch mouth

Name if gopal against that we have expression expression should be in key value format

When I am calling I always call the name: gopal all the three expressions will come or not

Instance_type: name

VM1: t2.micro
VM2: t3.micro
VM3: t3.large

Calling only instance_type

Concatenation of multiple values using local block

Output block

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Once I created the vpc and subnet. How do we confirm that the vpc and subnet is created or not?

Go to aws console then verify the same.

You an new developer I don't give you access to aws console. When we do remote state your state file will be stored in aws s3 bucket.

Input variables and local expression

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1. Where we can use the input variables
 2. Resource start with resource block
 3. Output start with output block
 4. Variables start with variables block
- Once you declare an variables to call a variable var.nameofthe variables.

Subnet and az

Tuesday, September 1, 2026

12:46 PM

1. We have created subnet application another for database.

2. We want to make our environment ha

Inside a region we have multiple az. Availability zone

Inside a region we have multiple data center they are called as az

For ha we should always use multiple data center

I want my subnet need to be created in two different az.

It will create the subnet in which ever is the best data center

I can create multiple subnet this subnet should be spread across multiple az data center

When you want to do similar kind of work

I used to forget the grocery list given by mom tea sugar and milk

Tea and sugar forget about milk. Again go and get the milk.

How to avoid this.

Create a list. Shop I will read the list. That way we will need not do repeat

If I have a list of cidrblock and az. I can use a loop for loop read the list and create the resource

Whenever there is an repeation of the task here we want to create multiple subnet or similar kind of vm will be created

This can be fulfill using loops

But loops required list of items

List of items

Tuesday, September 1, 2026 2:09 PM

What is a list. Collection of items are always there in the list.

When we go for shopping do we make list.

Midwest in us. County has a population of around 100 people. Quite cheper. Messuri. 40 miles 80 km.

We don't have any grocery store near by 80km. Once in a week an get the grocery for me and my family.

Fruit and Vegetables

Apples bananas strawberry

Staples

Oil

Dairy and refgerated.

#this list need to be arranged properly

grocery list

Kid1

Name: yesha

Fruit: apple

Quantity: 3

Is_snack: false

Preference: must have

Kid2

Name: Chrizeal

Fruit: orange

Quantity: 4

Is_snack: false

Preference: must have

This type of variables is called as object where you can group multiple attributes of different data types like string.

How we are managing kids grocery list.

#when you go to store we need to loop through all this variables list of items

Using terraform we want to create multiple in different az. Before I create the loop first I need to create the list of object

>

```
for (int i = 1; i <= 5; i++) { #inside the for block we have condition
    System.out.println("Count is: " + i);
}
```

Terraform is not an programming language

For_each = var.subnet

For_each will collect all the object

To call those object from my list in the list we have object in key value format

Each.key each.value

IG rtb attach

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#we will create an internet gw attach the same with the vpc

#we will create custom rtb attach with my vpc

#we will attach internet gw to the rtb

#we will attach all subnet to the custom rtb

Vm

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My networking is ready

We need to create few more things before I launch my vm

1. Security group where I ll open port 80 443 and 22
2. We need to get the image information : image
3. Custom script or startup script
4. Vm

Data block

Tuesday, September 1, 2026 4:12 PM

We are going to learn about data block

This is used for calling resources in the cloud which is created outside terraform.

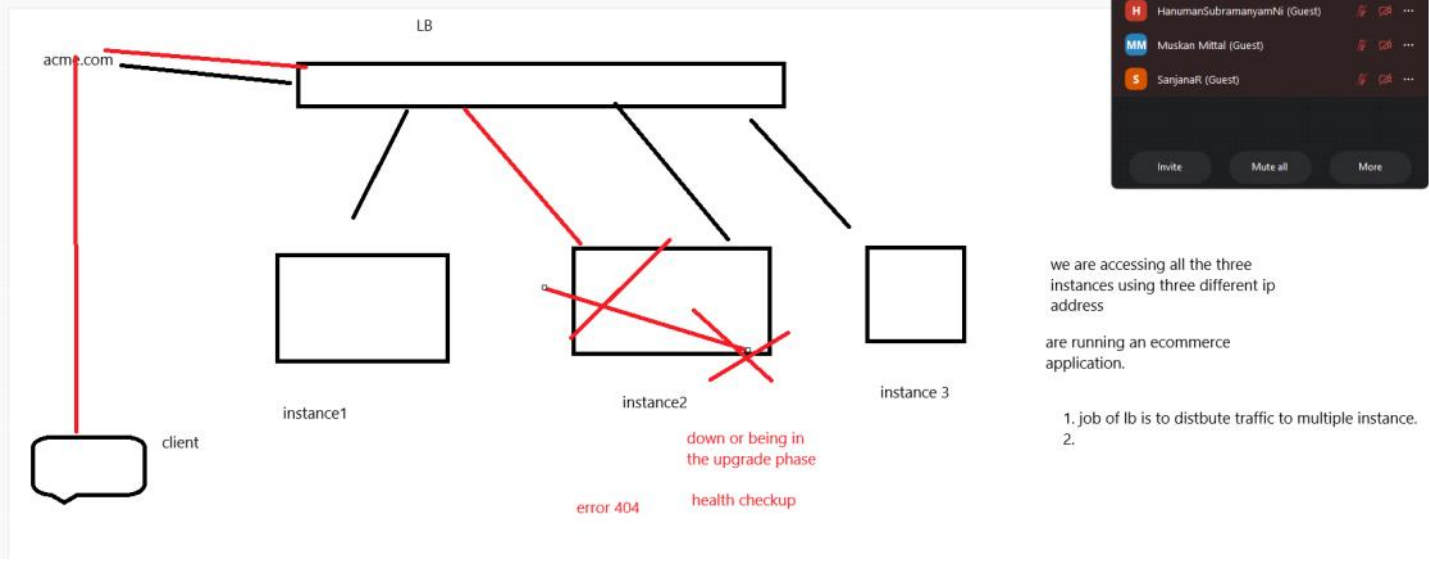
Image

Instance

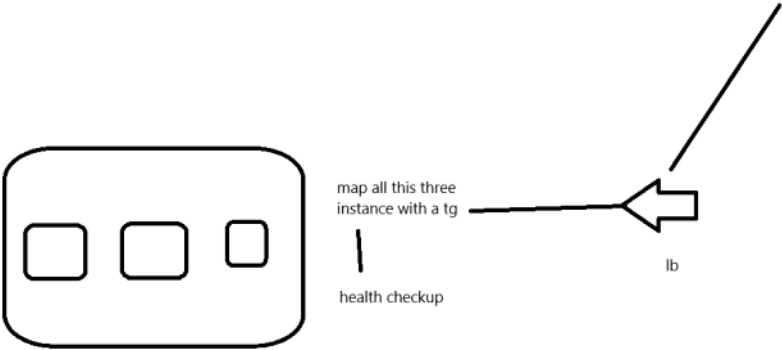
Wednesday, September 2, 2026 10:11 AM

- 1. Currently we have two subnet in two different az data center.
 - 2. I want to launch two different instance in two different az
 - 3. I also want that the size of the instance in different will be different in az1 t2.micro az2 t3.micro
- How to do the same.

Currently there are three instance and we are accessing those three instance using three different public ip.



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Screen clipping taken: 9/2/2026 11:44 AM

Lb

Wednesday, September 2, 2026 11:44 AM

1. We need to create a sg : to create the sg we don't need to write down the code from scratch we already have the instance sg we need to just modify it
2. Then we need to create a target group
3. Lb

Next we will create target group with this target group what we need to attach instance id

Statefile

Wednesday, September 2, 2026 2:26 PM

Currently the state file is stored in your system which contains the information about the resource creation in aws.

From day 1 till today we make changes

Terraform

Terraform destroy

Vpc and subnet ig

We make changes and apply the changes in the cloud those changes are being reflected inside your state file.

The main purpose of the state file is to understand terraform configuration and aws actual infrastructure.

When I am storing the state file in my laptop which has a read write storage.

Terraform apply : will go into terraform.tfstate

Terraform destroy: the older statefile will go inside terraform.tfstate.backup file

I got a new request to change the instance type from t2 micro to t3 micro.

We modify the instance so terraform apply so we lose the previous state

They want to see what you have done previously.

Read write storage

Object storage

It is a kind of storage which is read only. If you enable versioning in the storage .

I upload the same filename multiple times it will create a new version with timestamp

Amazon s3 simple storage service to store data

Inside the storage device if I enable versioning

Daily backup and want to automate the same. Backup file name will be same-upload in s3 bucket. It will not overwrite with the same filename

Backup:11:11:11

Whatever changes I have in the code a month back also we can check the same.

In aws we will create an s3 bucket. And store the statefile inside the bucket.

Version control system

Wednesday, September 2, 2026 3:52 PM

You are write code.

What you have written in first day

What you have today

You want to revisit the history.

You need a version control system.

We required a version control using the same I can easily manage my code. Git

Terraform cloud terraform module

You need a version control system to manage the same.

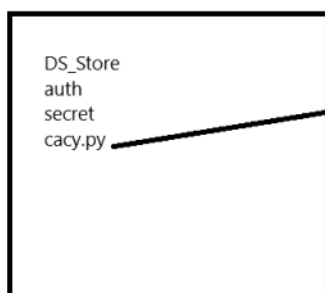
You client request to build infra using terraform. You write down the code. - will you apply it in client system ? Github which is the remote repo will be used

Git is the client which you use in your system

Github is the remote server where we upload the code for colloboration. Bitbucket azure repo gitlab.

The command in your local system will be same it don't change

projectx



share my application file.
git add cacy.py

this will move your file to an area
called as staging area



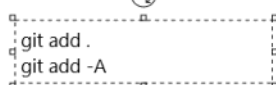
git commit -m "working on calcy app"



git comit will create a version of my file all
the files which is there in the staging area

git init- git will make this as
working directory

git add calcopy



git commit -m "added negative function"

Screen clipping taken: 9/2/2026 4:49 PM

Recap

Thursday, September 3, 2026 9:43 AM

We learn about version control system

1. Git workflow
2. Git init
3. Git add
4. Git commit

So every commit will create a version for you . It will keep all the changes.

We have committed two times.

We want to completely ignore some of the files. Git should stop tracking those files. Api key authentication or exe file binary I don't want to share lots of temp file or txt file generated I want to ignore the same completely

To do the same we need to create a file called as .gitignore inside file I need to put those file information this

Branches in git

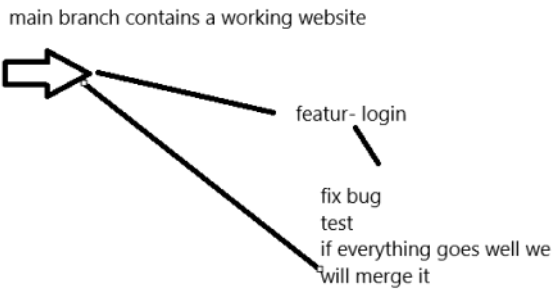
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Git branch is an independent line of development . It lets you make changes without disturbing the stable code on another branch usually main.



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for eg. imagine your main branch contains an working website. and you need to add an login page instead of changing the main directly you can create a seprate branch



Screen clipping taken: 9/3/2026 11:03 AM

n vergina
ohio

second region

hyd

main branch



vpc,subnet,ig,routet
atable vm lb
creation

deployed in aws
Mumbai location

change the name of the region
Mumbai to hyd

there are people in Chennai
they are complaining of the
latency in application
how to avoid this latency

Screen clipping taken: 9/3/2026 11:08 AM

Terraform cloud

Thursday, September 3, 2026 12:02 PM

Till now we have worked with terraform cli.

The problem with terraform cli is that your work will not be visible to others

We work in collaboration or team.

Silos always fail on the goal. When you work on team you can achieve the goal easily.

In evening night shift you will have a different person working on the same project. If the other team member wants to see what you have done earlier changes and what the new requirement they have received it will be very difficult to manage

Terraform cloud will provide a central platform.

Team collaboration will be easy

Every one can see the state file

Variables can be managed centrally

Whatever run history is there others can see

Audit trail

Policy

Rbac

Cicd

Continuous integration and continuous deployment.

I make changes in the code. Automatically the changes should be applied to the cloud service provider immediately.

Used to sell only veg product. They are in the market for past 12 years. Few days back they have shut down their operation.

When you are in faster deployment. In ecommerce add a feature of discount on tv product 40% discount declared. When people want to purchase they need to get the discount.

Before the deployment check has been created

Suddenly this month I got an AWS bill of 5 lakh but I got 15 lakh bill someone has launched an extra large instance with huge storage. Control so that I just allow only required instance type storage size 100 Gb

Linting

Thursday, September 3, 2026 4:37 PM

We are writing code this code need to match some quality. Is my code run into production we need to do quality testing to identify potential problem without running the application or creating infrastructure

Lint test: checking my code for mistakes bad practices inconsisten formation potential problem before execution.

Terraform validate.

Tflint tool which will do deep dive in your potential issue of the code

Some configuration issue

Terraform version

Required provider deprecated terraform syntax or feature

We declare a varialble but I am not using

Suspicious configuration patter

recap

Friday, September 4, 2026 9:36 AM

Git branching and how this branches will help us to manage your infra for dev test prod environment.
Security feature how to secure your code for vulnerability.

today

Friday, September 4, 2026 9:43 AM

Tfvars

Workspace in terraform

Modules in terraform using git how to manage this modules

###all this three concept is related to resuability

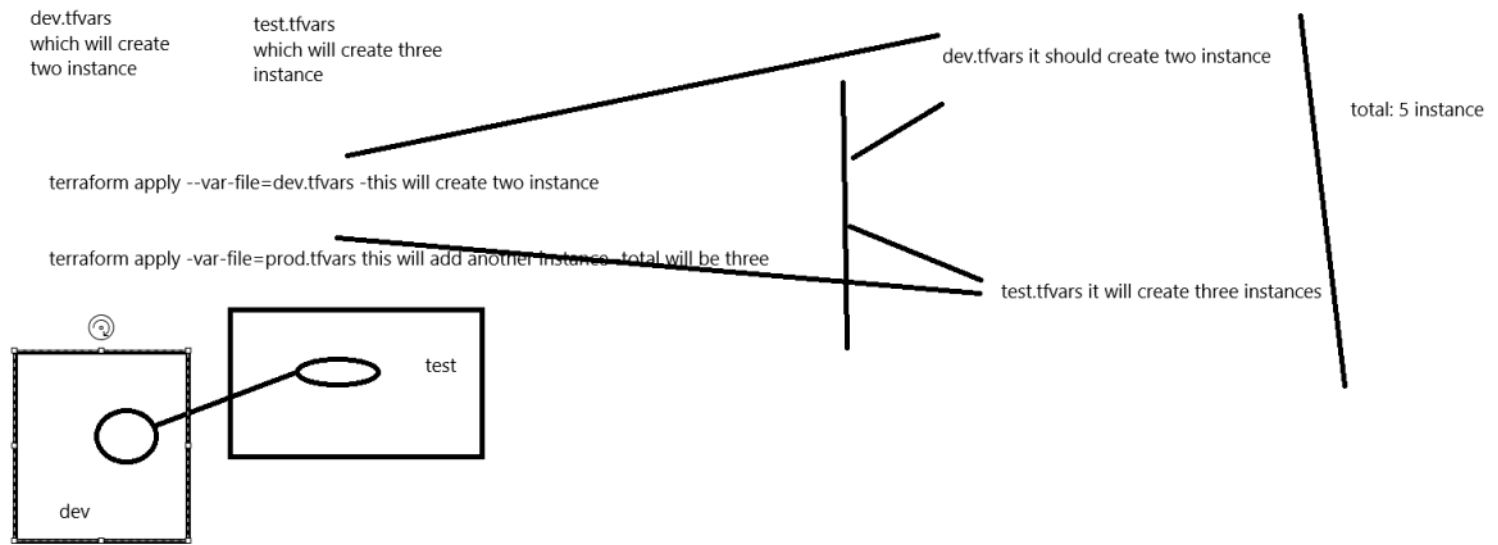
tfvars

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Workspace

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Using the workspace we can create multiple environment on the same folder



Screen clipping taken: 9/4/2026 11:29 AM

Statefile for workspace

Friday, September 4, 2026 12:06 PM

We have created two different dev another is test.

How the state file is managed by terraform using workspaces

In your s3 bucket automatically it create an ev folder inside that according to the name of your workspace it is going to create the state file

You want to deploy same infra in multiple region also

Modules in terraform

Friday, September 4, 2026 12:22 PM

Your client sit in overseas location. It is his wish.

You are an service provider to acme client.

They always give you request to create resources using terraform in the cloud.

The client is in payment gw or running medicine stores

PCI DSS or HIPAA GDPR Norms.

They want that whenever you create resources in the cloud you need to follow all the compliance or work according to company norms.

They implement some ai tools. What it do internally no one knows. Learning model. Bad things bad can happen.

Zero trust security in our resources in the cloud.

We create terraform code--we upload the same in github.

Two team member

1. Coder for IAC: as per client request they build the code. Then share the same using github. Client verify the same. Give an go ahead.
2. Who is going to create the resources using terraform. But they need to use the skeleton from the client github repo
3. Resource block you got it. Using the var file input values and do terraform apply the resource will be created.

Source: github.com/gopal/module/vpc

Cidr=11.0.0.0/16

Subnet 11.0.1.0/24

Terraform apply

Walmart:

Every month 40 people join the developer team. 10k application. In India they manage Easybuy

Every user need to added in the cloud service

Terraform script with the particular department

Add the user name. terraform apply inside azure ad automatically the user added.

Common job

Common task are there in your orgnization automate those common task with less friction. I don't need to write down anything from the scratch.

module directory structure

Friday, September 4, 2026 2:04 PM

How to create module directory structure.

From our existing code. We will split the reusable module into four logical folders.

Networking

Security-group

Compute

Alb

Each module will have the same standard terraform files

Main.tf : this will contains your resource definition

Variables.tf : this will contains variables

Output.tf : all the output values will be inside this file

##we will create this directory structure